



COLLEGE OF COMPUTING STUDIES, INFORMATION AND COMMUNICATION TECHNOLOGY

IT 311 – Advanced Database Systems

Laboratory Exercise No. 3: Normalization and Outlier Detection

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Course/Year/Section: BSIT 3A- NS	Date: September 8, 2025

Another important step of data cleansing is to identify unusual cases and remove them from the data set. In some situations, the outliers themselves might be the most interesting cases (detecting fraudulent credit card transactions, for example), but in most cases outliers are simply the result of an incorrect measurement and should be removed from the data set.

TASK 1: *Prepare the data (20 points)*

1. Drag the Titanic data into the process.
2. Add the operator **Select Attribute** and connect it.
3. Change the *Parameters* and remove Cabin, Lifeboat, Name, and Ticket number. Change the **attribute filter type** into **a subset**.
4. In the **select subset**, remove the four mentioned attributes above.



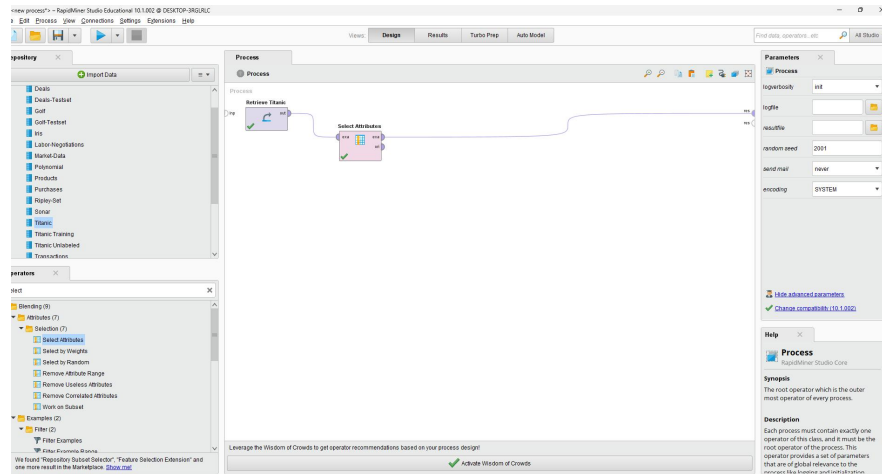
The result will be a data set only containing those columns we believe will contribute well to our outlier detection. We will use a distance-based outlier detection algorithm which calculates the **Euclidean distance between the data points** and marks those points which are farthest away from other data points as outliers. The Euclidean distance uses the distances between two data points for each individual attribute. Think about it: what is the effect on the distance if the attributes have different value ranges (one attribute between 0 and 5 and another attribute between 1 and 1000)? Attributes with larger values will contribute much more than those with smaller values. For this reason, we should ensure that all attributes are using similar value ranges. This transformation is called **Normalization**.



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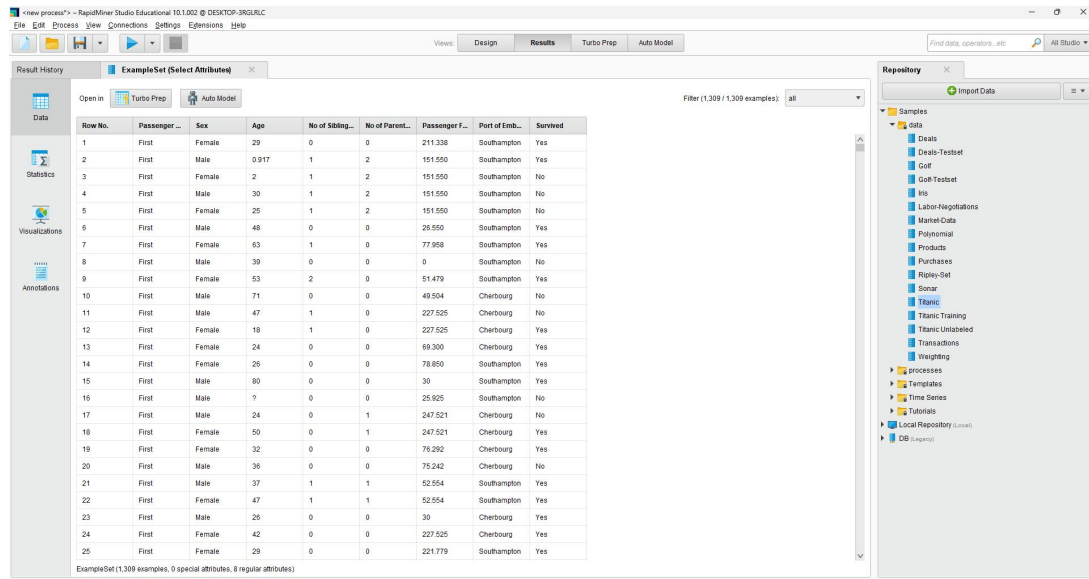
5. Take a screenshot of the process window and paste it here.

SCREENSHOT #1:



6. Run the process and take a screenshot of the remaining attributes.

SCREENSHOT #2:



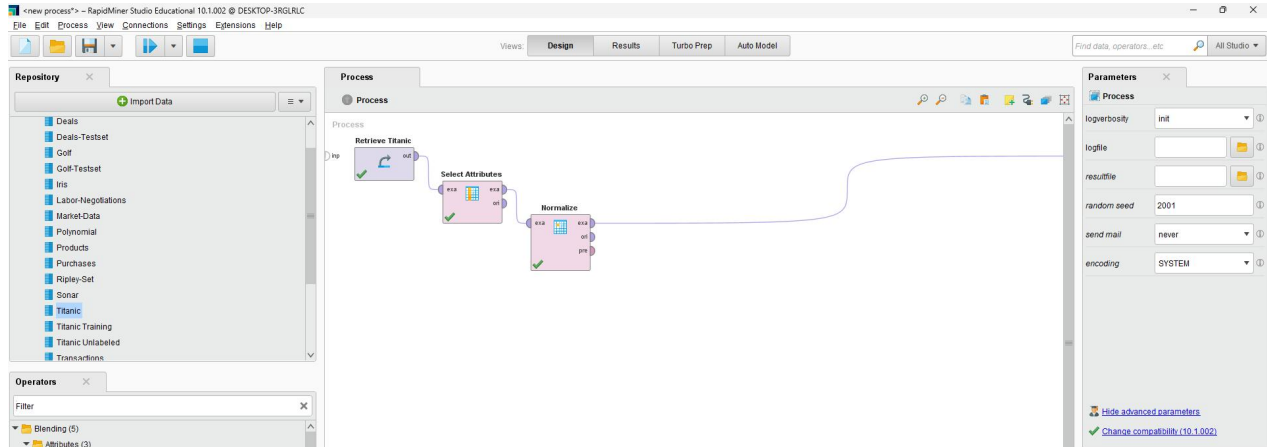


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TASK 2: *Normalized the Attribute Value Ranges (10points)*

1. Add the operator **Normalize** and connect it.
2. Take a screenshot of the process window.

SCREENSHOT #3:

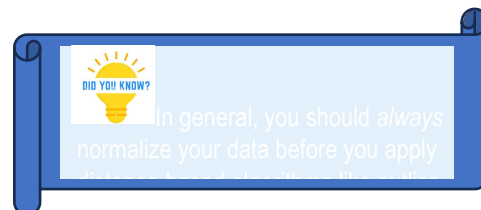


3. Run the process and take a screenshot of the result.

SCREENSHOT #4:

The screenshot shows the 'Result History' window in RapidMiner Studio, displaying the output of the 'Normalize' operator. The data is presented as a table with 25 rows and 10 columns. The columns are: Row No., Passenger..., Sex, Age, No of Siblings..., No of Parents..., Passenger F..., Port of Emb..., and Survived. The data represents the Titanic dataset, with each row corresponding to a passenger's record.

Row No.	Passenger...	Sex	Age	No of Siblings...	No of Parents...	Passenger F...	Port of Emb...	Survived
1	First	Female	29	0	0	211.336	Southampton	Yes
2	First	Male	0.917	1	2	151.550	Southampton	Yes
3	First	Female	2	1	2	151.550	Southampton	No
4	First	Male	30	1	2	151.550	Southampton	No
5	First	Female	25	1	2	151.550	Southampton	No
6	First	Male	48	0	0	25.050	Southampton	Yes
7	First	Female	63	1	0	77.958	Southampton	Yes
8	First	Male	39	0	0	0	Southampton	No
9	First	Female	53	2	0	51.479	Southampton	Yes
10	First	Male	71	0	0	49.524	Cherbourg	No
11	First	Male	47	1	0	227.925	Cherbourg	No
12	First	Female	18	1	0	227.925	Cherbourg	Yes
13	First	Female	24	0	0	69.309	Cherbourg	Yes
14	First	Female	26	0	0	78.850	Southampton	Yes
15	First	Male	60	0	0	30	Southampton	Yes
16	First	Male	?	0	0	25.925	Southampton	No
17	First	Male	24	0	1	247.521	Cherbourg	No
18	First	Female	58	0	1	247.521	Cherbourg	Yes
19	First	Female	32	0	0	70.292	Cherbourg	Yes
20	First	Male	36	0	0	75.242	Cherbourg	No
21	First	Male	37	1	1	52.054	Southampton	Yes
22	First	Female	47	1	1	52.054	Southampton	Yes
23	First	Male	26	0	0	30	Cherbourg	Yes
24	First	Female	42	0	0	227.925	Cherbourg	Yes
25	First	Female	29	0	0	221.779	Southampton	Yes



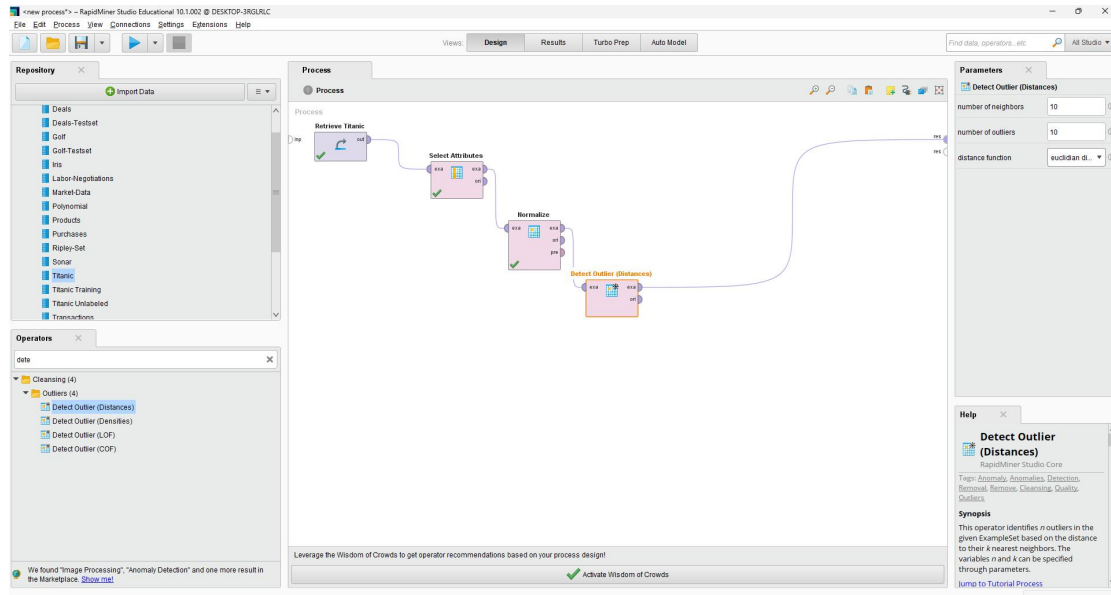


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TASK 3: Detect Outliers (10points)

1. Search for the operator **Detect Outlier (Distances)**, add it, and connect it to **Normalize**.
2. Take a screenshot of the process window.

SCREENSHOT #5:



3. **Run** the process and take a screenshot of the result.

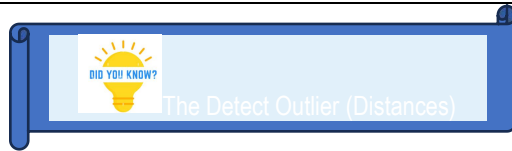
SCREENSHOT #6:

The screenshot shows the 'Results' tab in RapidMiner Studio, displaying the output of the 'Detect Outlier (Distances)' operator. The output is a table with 25 rows and 11 columns. The 'outlier' column contains the results of the detection, with 'false' indicating non-outliers and 'true' indicating outliers. The table is filtered to show 1,309 examples.

Row No.	outlier	Age	No of Sibling...	No of Parent...	Passenger f...	Passenger ...	Sex	Port of Emb...	Survived
1	false	-0.051	-0.479	-0.445	3.440	First	Female	Southampton	Yes
2	false	-2.910	0.481	1.866	2.285	First	Male	Southampton	Yes
3	false	-1.934	0.481	1.866	2.285	First	Female	Southampton	No
4	false	0.006	0.481	1.866	2.285	First	Male	Southampton	No
5	false	-0.339	0.481	1.866	2.285	First	Female	Southampton	No
6	false	1.257	-0.479	-0.445	-0.130	First	Male	Southampton	Yes
7	false	2.296	0.481	-0.445	0.863	First	Female	Southampton	Yes
8	false	0.533	-0.479	-0.445	-0.643	First	Male	Southampton	No
9	false	1.604	1.441	-0.445	0.351	First	Female	Southampton	Yes
10	false	2.853	-0.479	-0.445	0.313	First	Male	Cherbourg	No
11	false	1.188	0.481	-0.445	3.753	First	Male	Cherbourg	No
12	false	-0.824	0.481	-0.445	3.753	First	Female	Cherbourg	Yes
13	false	-0.408	-0.479	-0.445	0.696	First	Female	Cherbourg	Yes
14	false	-0.269	-0.479	-0.445	0.880	First	Female	Southampton	Yes
15	false	3.477	-0.479	-0.445	-0.064	First	Male	Southampton	Yes
16	false	?	-0.479	-0.445	-0.142	First	Male	Southampton	No
17	false	-0.408	-0.479	0.710	4.130	First	Male	Cherbourg	No
18	false	1.396	-0.479	0.710	4.130	First	Female	Cherbourg	Yes
19	false	0.147	-0.479	-0.445	0.831	First	Female	Cherbourg	Yes
20	false	0.425	-0.479	-0.445	0.810	First	Male	Cherbourg	No
21	false	0.494	0.481	0.710	0.372	First	Male	Southampton	Yes
22	false	1.188	0.481	0.710	0.372	First	Female	Southampton	Yes
23	false	-0.269	-0.479	-0.445	-0.064	First	Male	Cherbourg	Yes
24	false	0.841	-0.479	-0.445	3.753	First	Female	Cherbourg	Yes
25	false	-0.051	-0.479	-0.445	3.642	First	Female	Southampton	Yes



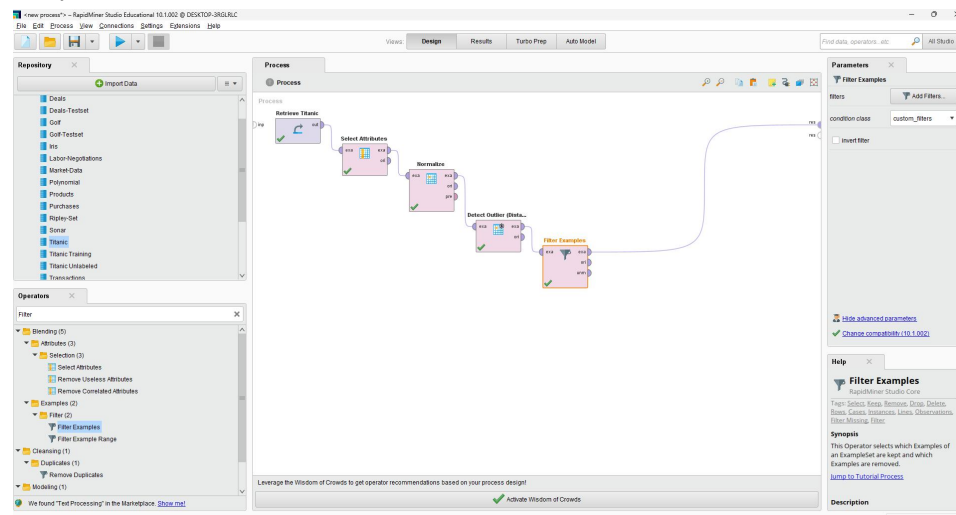
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TASK 4: Remove Outliers from the example set. (10points)

1. Add **Filter Examples** to the process and connect it to the previous operator and the result port on the right.
2. In its **Parameters**, add a new filter with **Outlier, equals, and false** as values.
3. Take a screenshot of the process window.

SCREENSHOT #7:



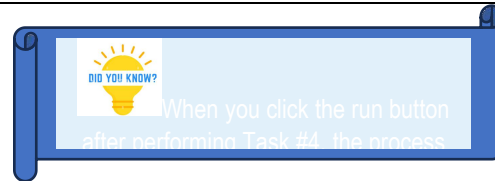
4. Run the process and take screenshot of the result.

SCREENSHOT #8:

Row No.	outlier	Age	No of Siblings	No of Parents	Passenger F.	Passenger L.	Sex	Port of Emb.	Survived
1	false	-0.061	-0.479	-0.445	3.440	First	Female	Southampton	Yes
2	false	-0.010	0.481	1.866	2.285	First	Male	Southampton	Yes
3	false	-1.934	0.481	1.866	2.285	First	Female	Southampton	No
4	false	0.008	0.481	1.866	2.285	First	Male	Southampton	No
5	false	-0.339	0.481	1.866	2.285	First	Female	Southampton	No
6	false	1.257	-0.479	-0.445	-0.130	First	Male	Southampton	Yes
7	false	2.298	0.481	-0.445	0.863	First	Female	Southampton	Yes
8	false	0.633	-0.479	-0.445	-0.843	First	Male	Southampton	No
9	false	1.604	1.441	-0.445	0.351	First	Female	Southampton	Yes
10	false	2.853	-0.479	-0.445	0.313	First	Male	Cherbourg	No
11	false	1.188	0.481	-0.445	3.753	First	Male	Cherbourg	No
12	false	-0.624	0.481	-0.445	3.753	First	Female	Cherbourg	Yes
13	false	-0.408	-0.479	-0.445	0.096	First	Female	Cherbourg	Yes
14	false	-0.269	-0.479	-0.445	0.880	First	Female	Southampton	Yes
15	false	3.477	-0.479	-0.445	-0.064	First	Male	Southampton	Yes
16	false	7	-0.479	-0.445	-0.142	First	Male	Southampton	No
17	false	-0.408	-0.479	0.710	4.139	First	Male	Cherbourg	No
18	false	1.386	-0.479	0.710	4.139	First	Female	Cherbourg	Yes
19	false	0.147	-0.479	-0.445	0.831	First	Female	Cherbourg	Yes
20	false	0.425	-0.479	-0.445	0.810	First	Male	Cherbourg	No
21	false	0.494	0.481	0.710	0.372	First	Male	Southampton	Yes
22	false	1.188	0.481	0.710	0.372	First	Female	Southampton	Yes
23	false	-0.269	-0.479	-0.445	-0.064	First	Male	Cherbourg	Yes
24	false	0.841	-0.479	-0.445	3.753	First	Female	Cherbourg	Yes
25	false	-0.061	-0.479	-0.445	3.542	First	Female	Southampton	Yes



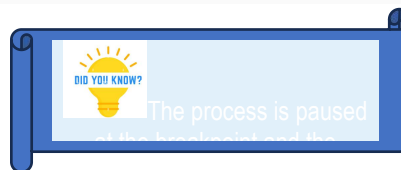
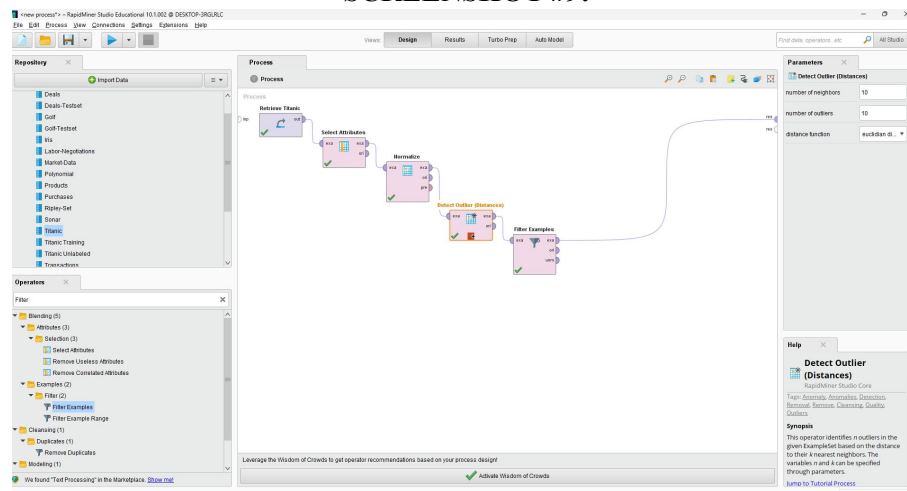
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TASK 5: Use Breakpoints to See Intermediate Results (10 points)

1. Go back to the design view.
2. Right click on **Detect Outlier (Distances)** and select **Breakpoint after** in the menu. Note that small icon appears at the bottom of the operator after you added the breakpoint. Take a screenshot of the process window.

SCREENSHOT #9:





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- Run the process again. Take a screenshot of the process window and the result.

SCREENSHOT #10:

new process...

RapidMiner Studio Educational 10.10.02 @ DESI2024-3962-ELC

File Edit Process View Connections Settings Extensions Help

Views

Design

Results

Setup Step

Auto Model

Find data, operations, etc

100 Results

ExampleSet (Detect Outlier (Distances))

Open in

Setup Step

Auto Model

Filter (1,308 / 1,308 examples)

all

Row No.	outlier	Age	No of Sibb...	No of Parent...	Passenger F...	Passenger...	Sex	Port of Emb...	Survived
1	-0.081	-0.479	-0.446	3.480	First	Female	Southampton	Yes	
2	-0.090	0.481	1.886	2.285	First	Male	Southampton	Yes	
3	-1.934	0.481	1.886	2.285	First	Female	Southampton	No	
4	-0.008	0.481	1.886	2.285	First	Male	Southampton	No	
5	-0.339	0.481	1.886	2.285	First	Female	Southampton	No	
6	1.257	-0.479	-0.446	-0.135	First	Male	Southampton	Yes	
7	2.268	0.481	-0.446	0.853	First	Female	Southampton	Yes	
8	0.933	-0.479	-0.446	-0.643	First	Male	Southampton	No	
9	1.054	1.441	-0.446	0.351	First	Female	Southampton	Yes	
10	2.853	-0.479	-0.446	0.313	First	Male	Cherbourg	No	
11	1.188	0.481	-0.446	3.753	First	Male	Cherbourg	No	
12	-0.024	0.481	-0.446	3.753	First	Female	Cherbourg	Yes	
13	-0.081	-0.479	-0.446	0.886	First	Female	Cherbourg	Yes	
14	-0.209	-0.479	-0.446	0.886	First	Female	Southampton	Yes	
15	3.477	-0.479	-0.446	-0.884	First	Male	Southampton	Yes	
16	-0.408	-0.479	-0.446	-0.142	First	Male	Southampton	No	
17	-0.408	-0.479	0.710	4.139	First	Male	Cherbourg	No	
18	1.395	-0.479	0.710	4.139	First	Female	Cherbourg	Yes	
19	0.147	-0.479	-0.446	0.851	First	Female	Cherbourg	Yes	
20	0.425	-0.479	-0.446	0.810	First	Male	Cherbourg	No	
21	0.494	0.481	0.710	0.372	First	Male	Southampton	Yes	
22	1.188	0.481	0.710	0.372	First	Female	Southampton	Yes	
23	-0.209	-0.479	-0.446	-0.884	First	Male	Cherbourg	Yes	
24	0.841	-0.479	-0.446	3.753	First	Female	Cherbourg	Yes	
25	-0.081	-0.479	-0.446	3.942	First	Female	Southampton	Yes	

ExampleSet (1,308 examples, 1 sparse attribute, 1 regular attribute)

Repository

Import Data

Examples

new data

ExampleSet

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TASK 6: Challenge (20 points)

- How would you change the process, so it finds 20 outliers instead of 10?
To find 20 outliers instead of 10, change the parameters from 10 into 20

Parameters

Detect Outlier (Distances)

number of neighbors: 10

number of outliers: 10

distance function: euclid...

- How can you change the process to only show outliers instead of removing them?

To only show outliers, we just need to change the filter to TRUE instead of false.

Create Filters: filters

Defines the list of filters to apply.

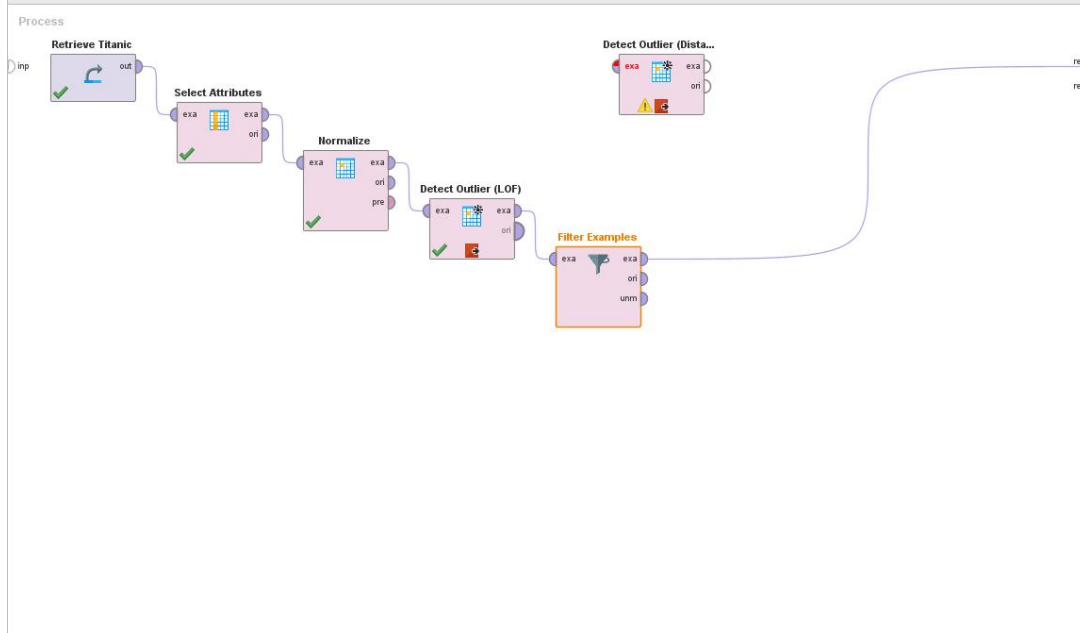
outlier equals false

- Replace the outlier detection operator with **Detect Outlier (LOF)** and a breakpoint after this operator before you execute. What is the difference to before?

The difference is that the distance-based method marks the outliers using global distance from all points, while LOF identifies outliers based on density difference, so the detected outliers may change depending on the structure.

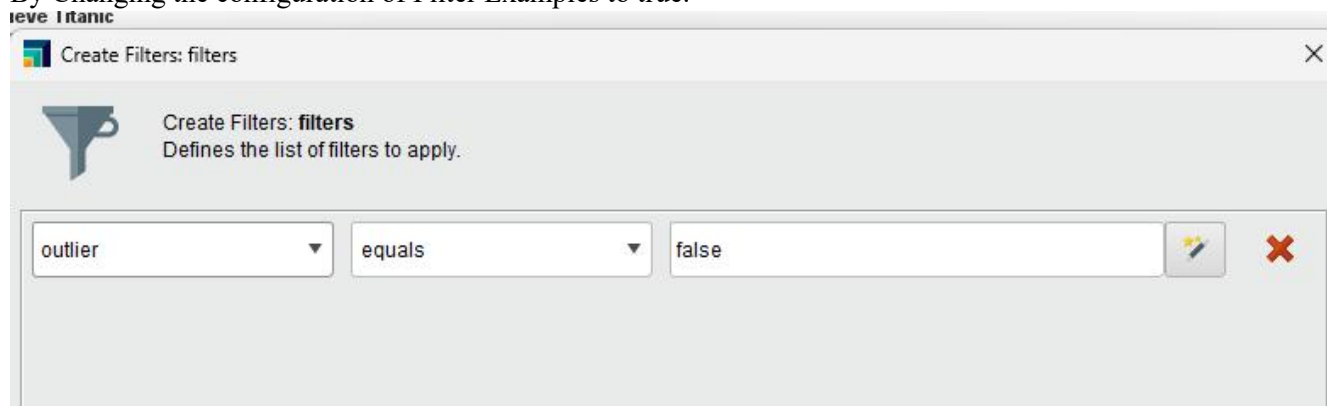


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4. How do you now need to change the filter to only keep the top outliers?

By Changing the configuration of Filter Examples to true.



IF Using Detect Outlier, change the range or threshold:



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Retrieve Titanic Detect Outlier (Dista...

Create Filters: filters

Create Filters: filters
Defines the list of filters to apply.

outlier = 1.5

Open in Turbo Prep Auto Model

Filter (1,309 / 1,309 examples): all

Row No.	outlier	Age	No of Sibling...	No of Parent...	Passenger F...	Passenger ...	Sex	Port of Emb...	Survived
1	1.030	-0.061	-0.479	-0.445	3.440	First	Female	Southampton	Yes
2	1.337	-2.010	0.481	1.866	2.285	First	Male	Southampton	Yes
3	1.364	-1.934	0.481	1.866	2.285	First	Female	Southampton	No
4	1.167	0.008	0.481	1.866	2.285	First	Male	Southampton	No
5	1.180	-0.339	0.481	1.866	2.285	First	Female	Southampton	No
6	1.356	1.257	-0.479	-0.445	-0.130	First	Male	Southampton	Yes
7	1.265	2.298	0.481	-0.445	0.863	First	Female	Southampton	Yes
8	1.614	0.633	-0.479	-0.445	-0.643	First	Male	Southampton	No
9	1.267	1.604	1.441	-0.445	0.351	First	Female	Southampton	Yes
10	2.307	2.853	-0.479	-0.445	0.313	First	Male	Cherbourg	No
11	1.180	1.188	0.481	-0.445	3.753	First	Male	Cherbourg	No
12	1.342	-0.824	0.481	-0.445	3.753	First	Female	Cherbourg	Yes
13	0.925	-0.408	-0.479	-0.445	0.696	First	Female	Cherbourg	Yes
14	1.056	-0.269	-0.479	-0.445	0.880	First	Female	Southampton	Yes
15	3.298	3.477	-0.479	-0.445	-0.064	First	Male	Southampton	Yes
16	0	?	-0.479	-0.445	-0.142	First	Male	Southampton	No
17	1.299	-0.408	-0.479	0.710	4.139	First	Male	Cherbourg	No
18	1.144	1.396	-0.479	0.710	4.139	First	Female	Cherbourg	Yes
19	1.022	0.147	-0.479	-0.445	0.831	First	Female	Cherbourg	Yes
20	1.135	0.425	-0.479	-0.445	0.810	First	Male	Cherbourg	No
21	1.297	0.494	0.481	0.710	0.372	First	Male	Southampton	Yes
22	1.078	1.188	0.481	0.710	0.372	First	Female	Southampton	Yes
23	1.119	-0.269	-0.479	-0.445	-0.064	First	Male	Cherbourg	Yes
24	0.976	0.841	-0.479	-0.445	3.753	First	Female	Cherbourg	Yes
25	1.056	-0.061	-0.479	-0.445	3.642	First	Female	Southampton	Yes

ExampleSet (1,309 examples, 1 special attribute, 8 regular attributes)